Fiscal Year:	FY 2020	**Task Last Updated:**	FY 12/02/2019
PI Name:	Steen, Paul H Ph.D.		
Project Title:	Capillary Flows and Draining in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	14853-5201	**Congressional District:**	23
Comments:	PI is deceased as of September 2020		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2017 Physical Sciences NNH17ZTT001N-17PSI-D: Use of the NASA Physical Sciences Informatics System – Appendix D
Start Date:	01/31/2019	**End Date:**	01/30/2021
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
Contact Monitor:	Hatch, Tyler	**Contact Phone:**	216.433.5073
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	NOTE (Ed., Mar 2019): This work is result of proposal by graduate student Joshua McCraney, Ph.D. candidate at Cornell University, who will be performing the work with advisor PI Dr. Paul Steen. NOTE: (Ed., October 2020): Dr. Olivier Desjardins will be the new Principal Investigator due to the passing of Dr. Steen in September 2020.		
COI Name (Institution):	McCraney, Joshua M.S. (Graduate Student PI: Cornell University)		
Grant/Contract No.:	80NSSC19K0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Capillary Flow Experiments (CFE) for Interior Corner Flows (ICF) were conducted aboard the International Space Station (ISS) and rely solely on video for high-quality quantitative data. We propose to digitally measure the still video images of the time-dependent interface height. Since container geometry is known, the meniscus heights can be used to determine volumetric drain rates. This data would then serve as both a benchmark for current theory on drain rates and a guide for extending capillary flow models to a diverse set of geometries and flow conditions. We also propose to utilize and extend a numerical tool to simulate 3D transient corner flows with and without topological change. To pursue any discrepancy between data and numerics, supportive ground-based experiments at micro-scales and low-gravity experiments conducted in the Dryden Drop Tower at Portland State University are proposed. Understanding the dynamics of capillary flows and responses of drain flows to various geometries offers design advantages for spacecraft waste-management systems, fuel control, hydration containment, and cryogenic flows. Capillary flows and drain problems are relevant to a broad range of two-phase flow technologies in low gravity, including 3D vane networks for positioning of fuels, an important problem concerning propellant liquid management aboard spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We highlight progress over the last six months, as NASA Physical Sciences Informatics (PSI) funding began mid-May 2019. Two outcomes of the proposed work are to develop a numerical tool to simulate low gravity fluid flow and to benchmark the numerical simulations and developed theory with International Space Station (ISS) flight data. To this end, we have mined the ISS database and identified ISS video data relevant to proposed fluid drain studies. Several observed videos exhibit complicated flow phenomena; further, the container geometry within the videos is complex, thus difficult to model. As a first step, we then analyze simpler flow geometries. We developed predictive flow models for these simpler geometries, and presented models at the American Physical Society Division of Fluid Dynamics (APS DFD) conference in November, 2019. The theoretical developments offer insight into fuel and cryogenic management aboard spacecraft and micro terrestrial systems. Such insight can assist design engineers manufacturing propellant management devices. We have begun extending theoretical models to the PSI geometries, but theory alone will not capture all the flow dynamics. In accordance with our proposal, we developed a numerical tool to model low gravity fluid motions. OpenFOAM, a computational fluid dynamics software, was selected to build the numerical tool since it is open-source software, and thus easily accessible to the scientific community at large, who can access our code at the PSI database at the conclusion of our investigations. Critical fluid dynamics necessary to accurately simulate the experiments cannot be measured/viewed from the video data alone: the PSI data was filmed from a side view, inhibiting contact line measurements. In accordance with the proposal, we manufactured supportive terrestrial experiments with a view perspective to measure the contact line. These design measurements are guided by developed theory and OpenFOAM simulations; then these experimental results will be compared to developed theory and numerical simulations.
Bibliography Type:	Description: (Last Updated: 09/04/2025)
Abstracts for Journals and Proceedings	McCraney J, Steen PH, Bostwick JB. "Low-G inertial-capillary meniscus motions in a channel." 72nd Annual Meeting of the American Physical Society's Division of Fluid Dynamics (DFD), Seattle, WA, November 23-26, 2019. Bulletin of the American Physical Society. 2019 Nov;64: http://meetings.aps.org/Meeting/DFD19/Session/G08.4 , Nov-2019